

K-Nearest Neighbour (KNN) classifier -

Sepal length	Sepal width	Species
5.3	3.7	Setosa
5.1	3.8	Setosa
7.2	3.0	Virginica
5.4	3.4	Setosa
5.1	3.3	Setosa
5.4	3.9	Setosa
7.4	2.8	Virginica
6.1	2.8	Versicolor
7.3	2.9	Virginica
6.0	2.7	Versicolor
5.8	2.8	Virginica
6.3	2.3	Versicolor
5.1	2.5	Versicolor
6.3	2.5	Versicolor
5.5	2.5	Versicolor

Find

Sepal length	Sepal width	Species
5.2	3.1	?

Given, $K = 5$ (based on 5 nearest neighbors)

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find euclidean distance,

$$D(x, y) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

find nearest neighbor

with input (5.2, 3.1)

Sepal length	Sepal width	Species	Distance
5.3	3.7	Setosa	0.608
5.1	3.8	Setosa	0.707
7.2	3.0	Virginica	2.002
5.4	3.4	Setosa	0.361
5.1	3.3	Setosa	0.224
5.4	2.9	Setosa	0.825
7.4	2.8	Virginica	2.220
6.1	2.8	Versicolor	0.949
7.3	2.9	Virginica	2.110
6.0	2.7	Versicolor	0.894
5.8	2.8	Virginica	0.671
6.3	2.3	Versicolor	1.360
5.1	2.5	Versicolor	0.608
6.3	2.5	Versicolor	1.253
5.5	2.5	Versicolor	0.671

Setosa → 3
Versicolor → 7

So, majority is Setosa, so, the given input belongs to class "Setosa".